

Data Analysis for the Titanic Dataset with Pandas and Matplotlib

What is Matplotlib?

Its main job is to take your data (like numbers in a list or a Pandas `DataFrame`) and turn it into a **static picture** (a plot or chart).

It's incredibly powerful and customizable. Almost any plot you can imagine, you can build with Matplotlib.

The Most Important Concept: `fig` and `ax`

`fig, ax = plt.subplots()`. This is the "object-oriented" way to plot, and it's the most flexible.

Let's break down that one line: `fig, ax = plt.subplots(figsize=(10, 6))`

- `plt.subplots()`: This is a function that creates two things for you:
 1. **`fig` (The Figure)**: This is the **entire window or page** your plot lives on. Think of it as the picture frame. You use `fig` to do things like save the plot to a file (e.g., `fig.savefig('my_plot.png')`).
 2. **`ax` (The Axes)**: This is the **actual plot** itself—the part with the data, labels, and ticks. Think of it as the canvas inside the frame. This is what you'll use 99% of the time. You call methods on `ax` to create the plot, like `ax.bar()`, `ax.scatter()`, and `ax.set_title()`.

Analogy: `fig` is the picture frame, `ax` is the canvas you draw on.

Setup: Loading the Titanic Data

You can load the Titanic Dataset directly from `Kaggle` into the pandas `DataFrame`.

```
import matplotlib.pyplot as plt
import pandas as pd
import kaggle

# Load the dataset
kaggle.api.authenticate()
kaggle.api.dataset_download_files('yasserh/titanic-dataset', path='data/', unzip=True)
df = pd.read_csv('data/Titanic-Dataset.csv')

# Let's see what columns we have
print(df.head())
```

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The Basic Plots

Here are the most common plots, what they're for, and how to make them with your `ax` object.

1. Bar Plot

- **What is it?** A plot that uses rectangular bars to compare values for different **categories**.
- **When to use it?** When you want to compare a numerical value (like a count or an average) across several distinct groups.

Titanic Example 01: Let's compare the number of survivors in each passenger class (`pclass`).

```
In [83]: import pandas as pd
df = pd.read_csv('Titanic-Dataset.csv')

In [84]: df
```

Out[84]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows × 12 columns

In [85]:

```
df['Age'] = df['Age'].fillna(df['Age'].median())
```

In [86]:

```
import matplotlib.pyplot as plt
```

In [87]:

```
# --- setup ---
fig, ax = plt.subplots(figsize=(8, 5))

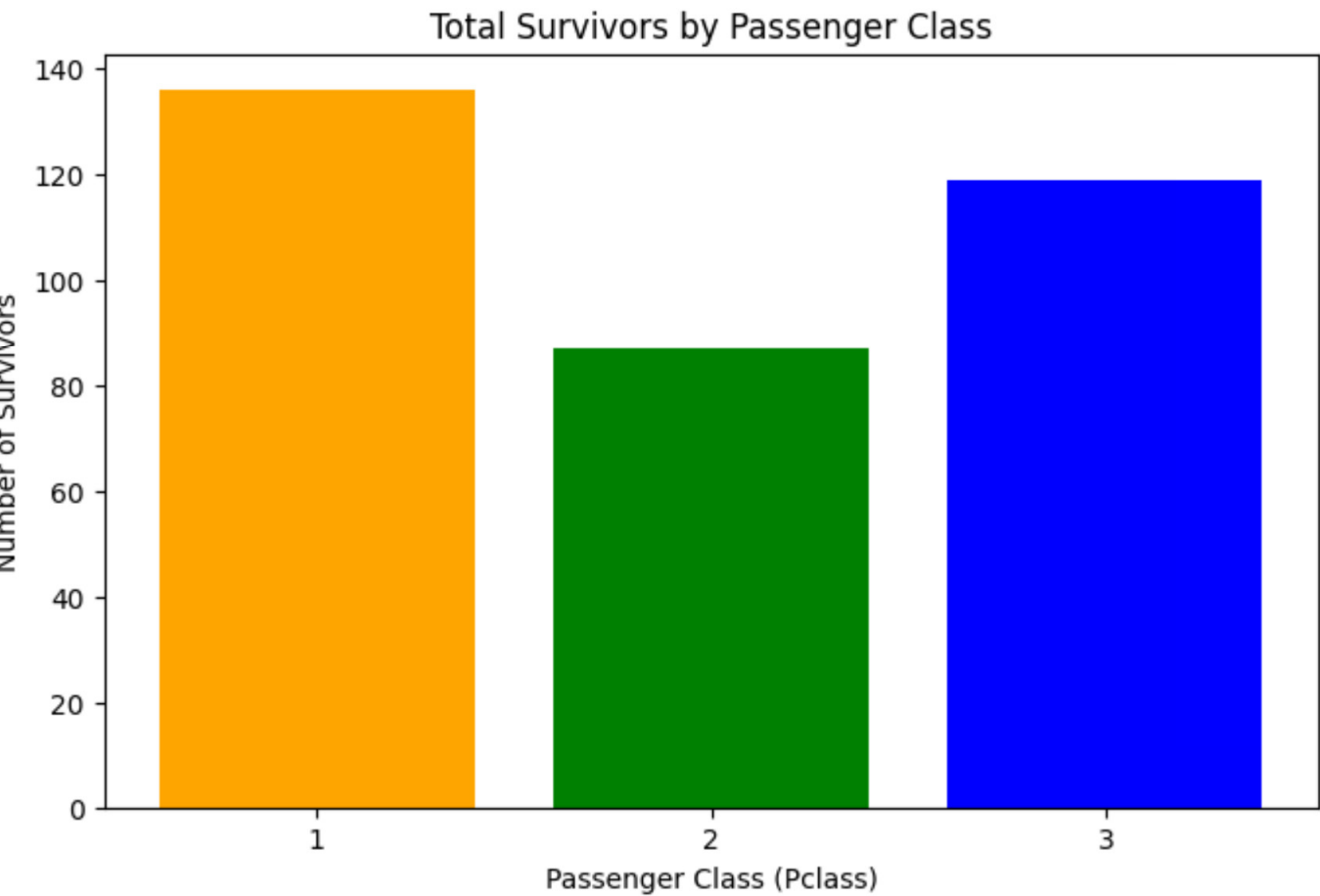
survivedByClass = df.groupby('Pclass')['Survived'].sum()

ax.bar(survivedByClass.index, survivedByClass.values, color = ['orange', 'green', 'blue'])

ax.set_title('Total Survivors by Passenger Class')
ax.set_xlabel('Passenger Class (Pclass)')
ax.set_ylabel('Number of Survivors')
ax.set_xticks([1, 2, 3]) # set clean ticks on the x-axis
```

Out[87]:

```
[<matplotlib.axis.XTick at 0x1981c5d1590>,
 <matplotlib.axis.XTick at 0x1981c5d3c50>,
 <matplotlib.axis.XTick at 0x1981c628410>]
```



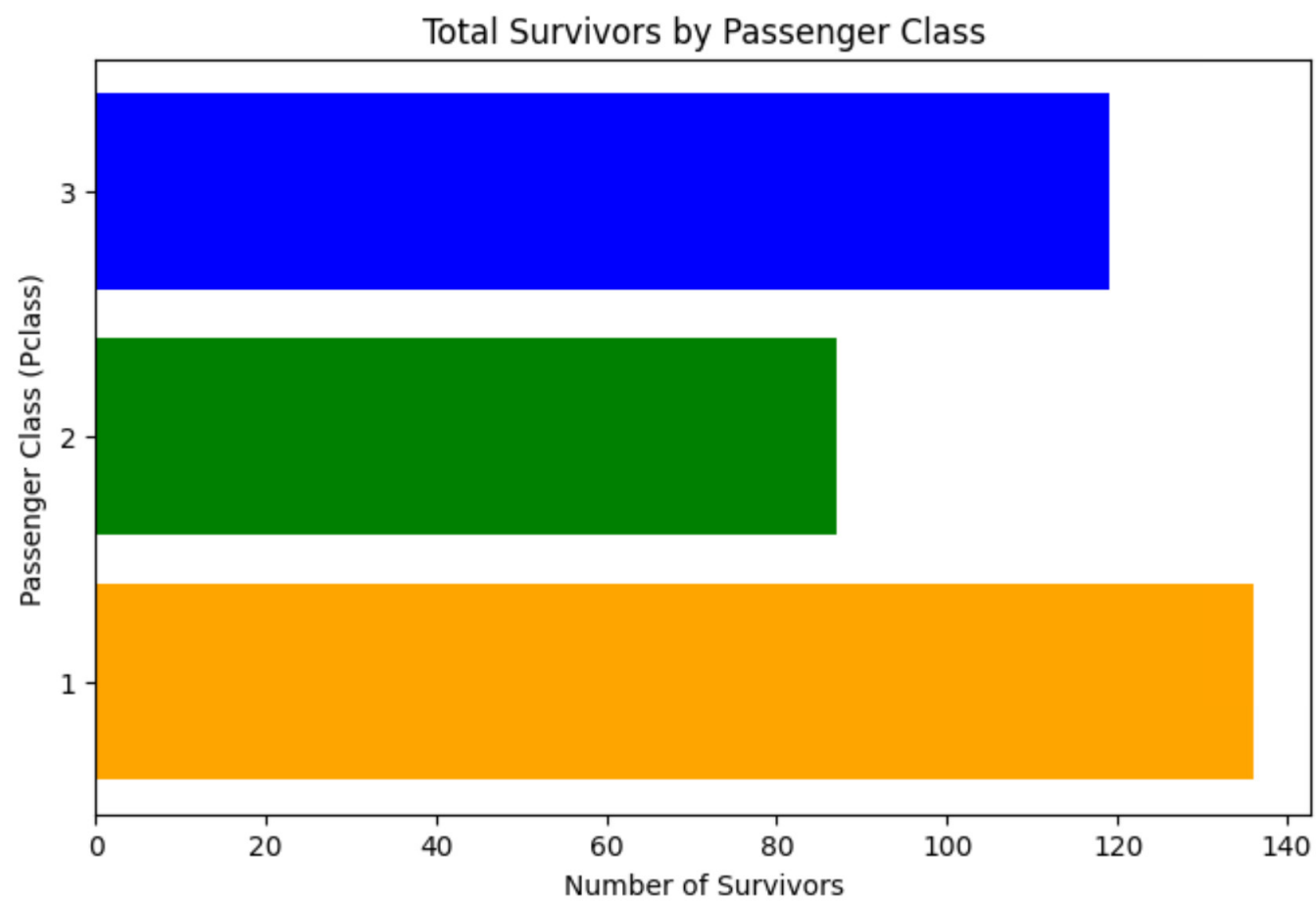
In [88]:

```
fig, ax = plt.subplots(figsize=(8, 5))
ax.barh(survivedByClass.index, survivedByClass.values, color = ['orange', 'green', 'blue'])

ax.set_title('Total Survivors by Passenger Class')
ax.set_xlabel('Number of Survivors')
ax.set_ylabel('Passenger Class (Pclass)')
ax.set_yticks([1, 2, 3]) # set clean ticks on the x-axis
```

Out[88]:

```
[<matplotlib.axis.YTick at 0x1981c64df90>,
 <matplotlib.axis.YTick at 0x1981c64fed0>,
 <matplotlib.axis.YTick at 0x1981c698690>]
```



Common Parameters for `ax.bar()` :

- `x` : The categories (our `pclass` index: `[1, 2, 3]`).
- `height` : The numerical value for each bar (our survivor counts).
- `color` : Sets the color of all bars (e.g., `color='skyblue'`).
- `edgecolor` : Sets the color of the bar outlines (e.g., `edgecolor='black'`).
- `label` : A name for this data, used if you call `ax.legend()`.

Pro-tip: Use `ax.barh()` for a **horizontal bar chart** (you'd use `y` and `width` parameters instead).

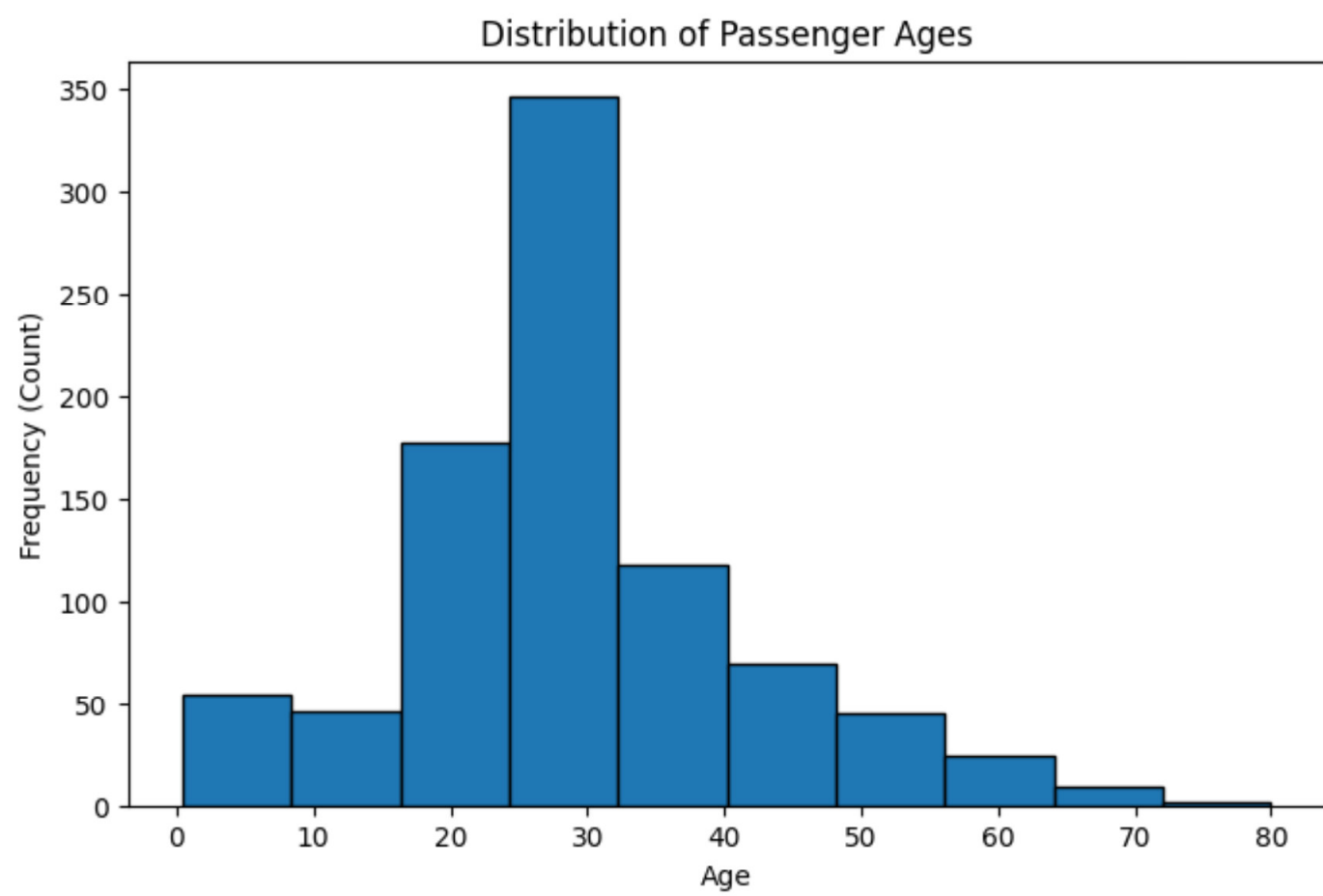
2. Histogram

- **What is it?** A plot that shows the **distribution** (or frequency) of a *single continuous* variable. It groups data into "bins" (buckets) and plots the count for each bin.
- **When to use it?** When you want to see the *shape* of your data. For example: "Are most passengers young? Are high fares common or rare?"

Titanic Example 02: Let's look at the **distribution of passenger ages**.

```
In [89]: fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(df['Age'], edgecolor = 'black')
ax.set_title('Distribution of Passenger Ages')
ax.set_xlabel('Age')
ax.set_ylabel('Frequency (Count)')
```

```
Out[89]: Text(0, 0.5, 'Frequency (Count)')
```

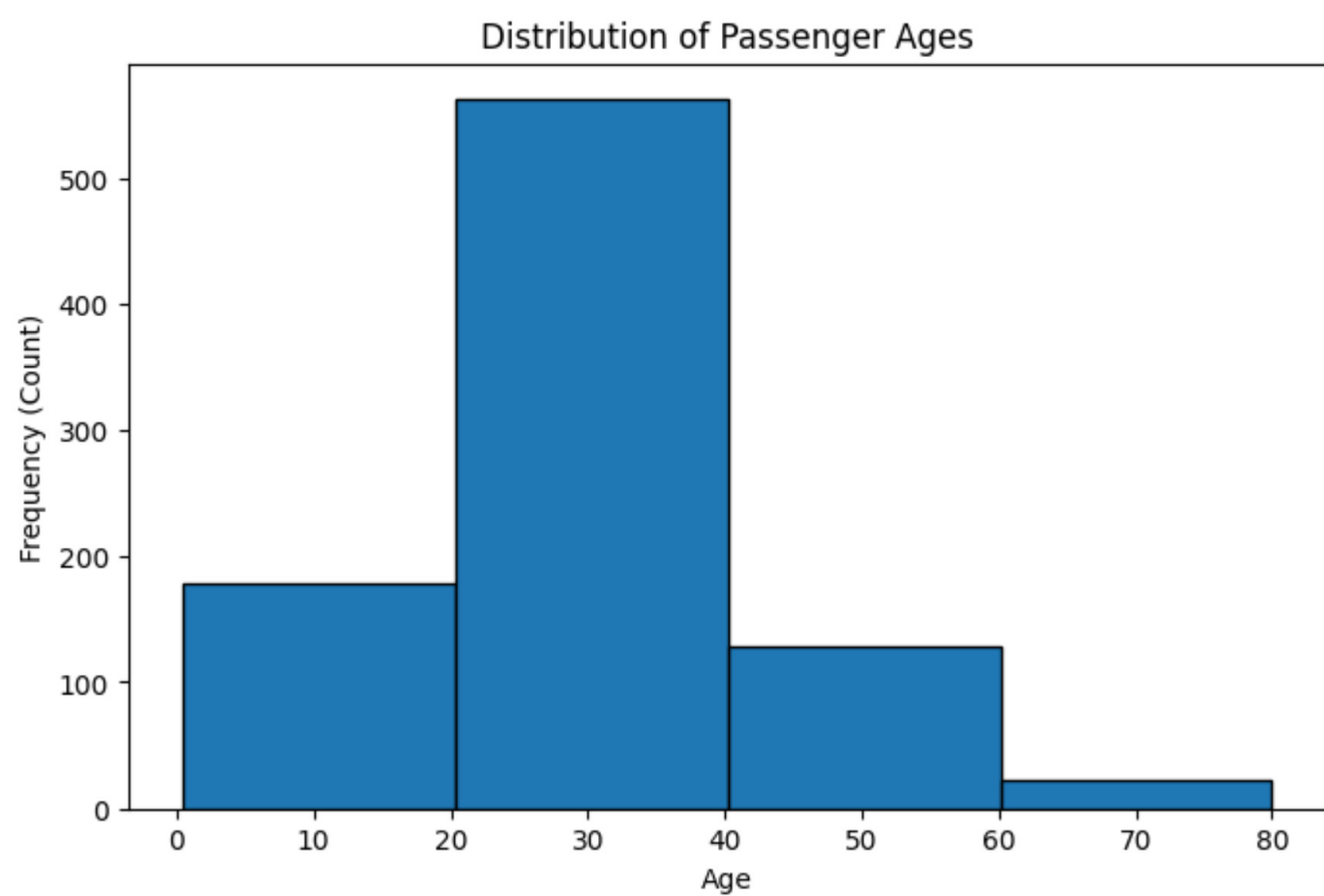



```
In [90]: fig, ax = plt.subplots(figsize=(8, 5))

ax.hist(df['Age'], bins= 4, edgecolor = 'black')

ax.set_title('Distribution of Passenger Ages')
ax.set_xlabel('Age')
ax.set_ylabel('Frequency (Count)')
```

Out[90]: Text(0, 0.5, 'Frequency (Count)')



Titanic Example 03: Distribution of passenger ages of survivals.

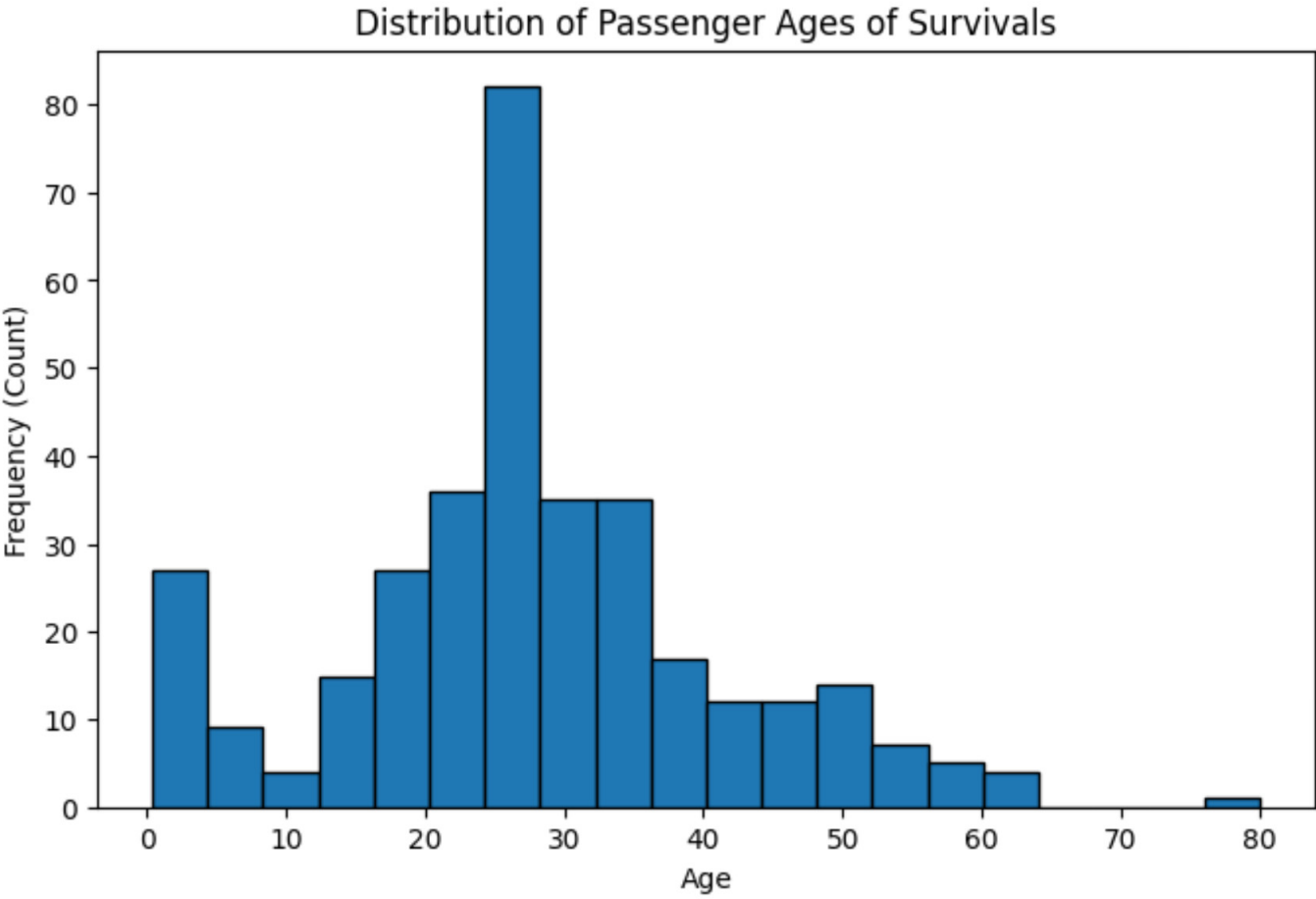
```
In [91]: fig, ax = plt.subplots(figsize=(8, 5))

ax.hist(df[df['Survived']==1]['Age'], bins= 20, edgecolor = 'black')

ax.set_title('Distribution of Passenger Ages of Survivals')
ax.set_xlabel('Age')
ax.set_ylabel('Frequency (Count)')
```

Out[91]: Text(0, 0.5, 'Frequency (Count)')

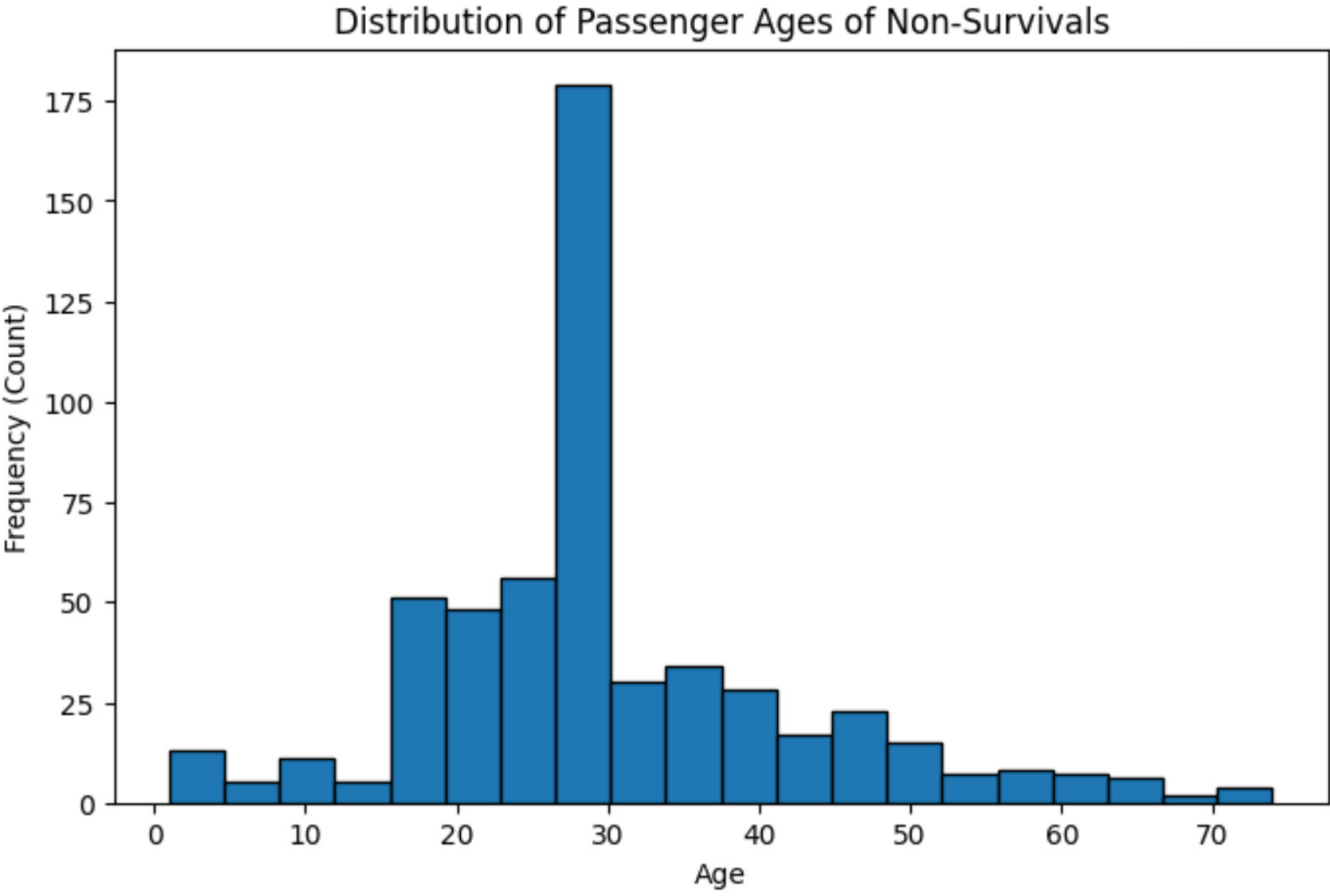
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Titanic Example 04: Distribution of passenger ages of Non-Survivals.

```
In [92]: fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(df[df['Survived']==0]['Age'], bins= 20, edgecolor = 'black')
ax.set_title('Distribution of Passenger Ages of Non-Survivals')
ax.set_xlabel('Age')
ax.set_ylabel('Frequency (Count)')
```

Out[92]: Text(0, 0.5, 'Frequency (Count)')

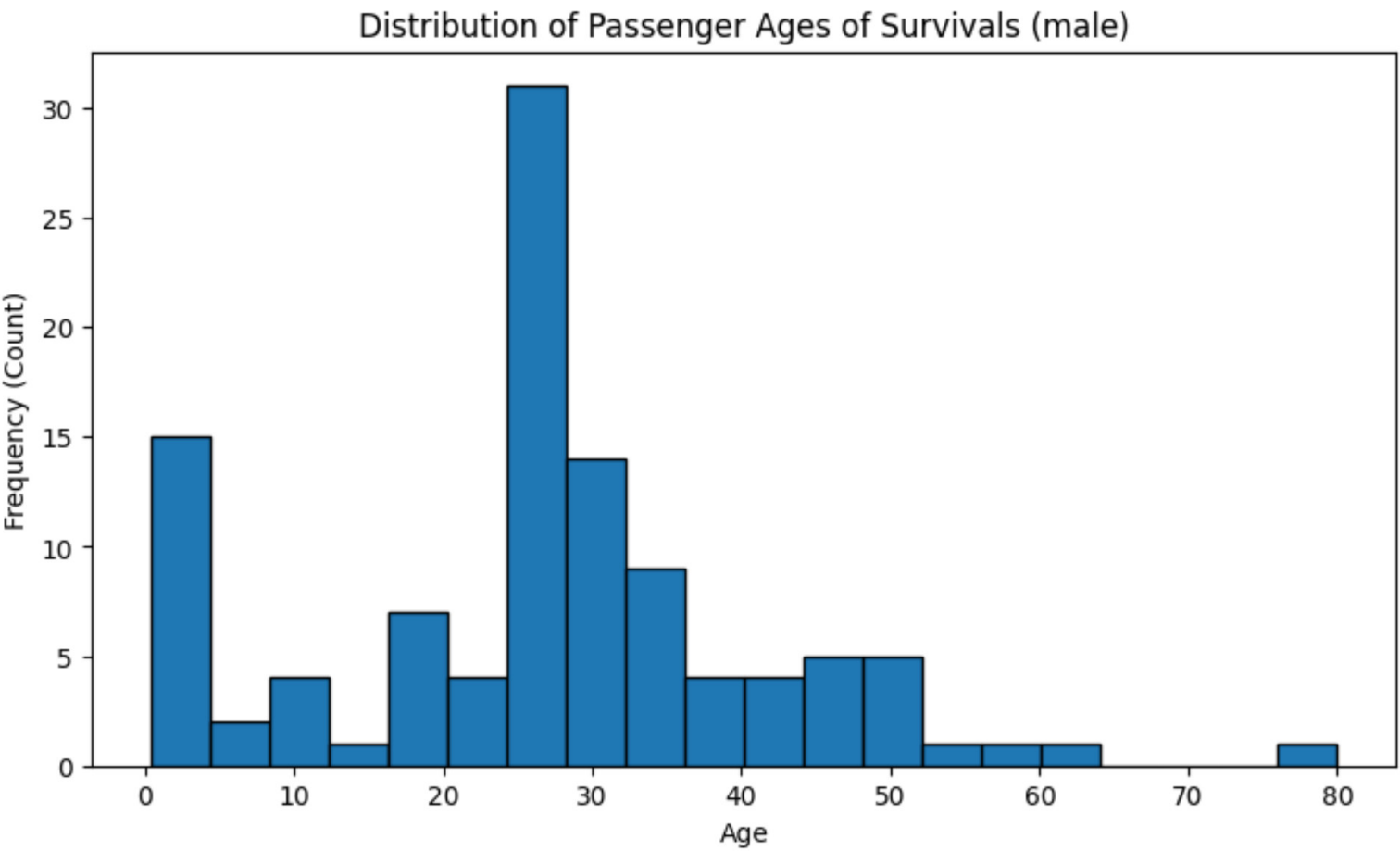


Titanic Example 05: Distribution of passenger ages of Survivals by Gender.

```
In [93]: fig, ax = plt.subplots(1,2,figsize=(20, 5))
ax[0].hist(df[(df['Survived']==1) & (df['Sex']=='male')]['Age'], bins= 20, edgecolor = 'black')
ax[0].set_title('Distribution of Passenger Ages of Survivals (male)')
ax[0].set_xlabel('Age')
ax[0].set_ylabel('Frequency (Count)')
ax[1].hist(df[(df['Survived']==1) & (df['Sex']=='female')]['Age'], bins= 20, edgecolor = 'black', color = 'pink')
ax[1].set_title('Distribution of Passenger Ages of Survivals (female)')
ax[1].set_xlabel('Age')
ax[1].set_ylabel('Frequency (Count)')
```

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Out[93]: Text(0, 0.5, 'Frequency (Count)')



Common Parameters for `ax.hist()` :

- `x` : The (single) array or list of data to plot.
- `bins` : This is the most important one! It's the number of buckets to create.
 - `bins=20` (Create 20 equal-width bins).
 - `bins=[0, 10, 20, 30, 60, 80]` (Define the exact bin edges).
- `color` : The fill color of the bars.
- `edgecolor` : The outline color of the bars.
- `alpha` : Transparency (from `0.0` to `1.0`). Useful for overlapping histograms (e.g., `alpha=0.7`).

3. Scatter Plot

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- **What is it?** A plot that shows the value of individual data points by placing a dot for each one.
- **When to use it?** To see the **relationship between two continuous variables**.

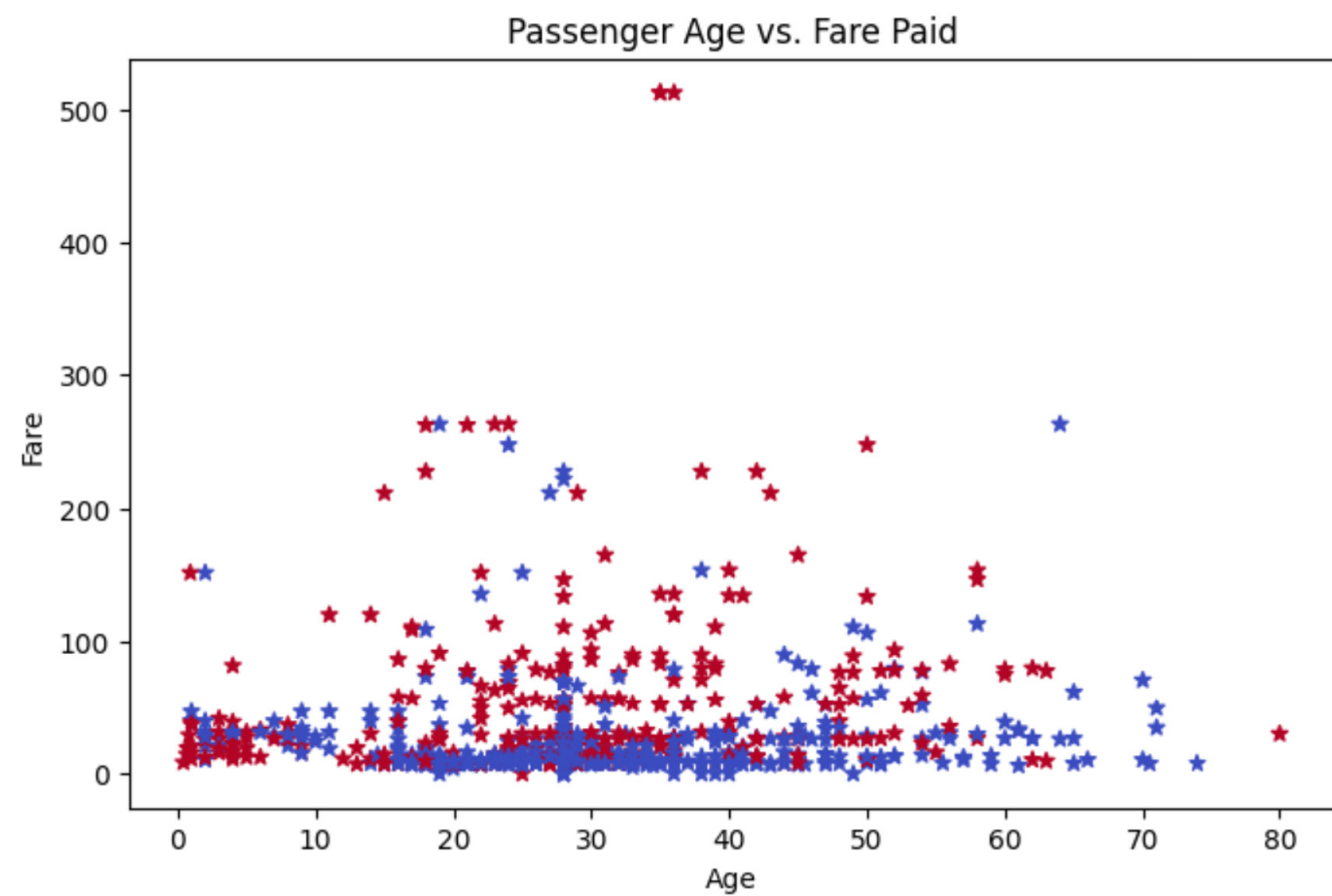
Titanic Example 06: Let's see if there's a relationship between a passenger's `age` and the `fare` they paid.

```
In [94]: fig, ax = plt.subplots(figsize=(8, 5))

ax.scatter(x=df['Age'], y=df['Fare'], c=df['Survived'], marker='*', cmap='coolwarm')

ax.set_title('Passenger Age vs. Fare Paid')
ax.set_xlabel('Age')
ax.set_ylabel('Fare')
```

Out[94]: Text(0, 0.5, 'Fare')



Common Parameters for `ax.scatter()` :

- `x` , `y` : The two variables you want to compare.
- `s` : The **size** of each marker.
- `c` : The **color**. This is powerful! You can set it to a third variable.
 - *Example:* `c=df['Survived']` would color the dots differently based on whether the passenger survived.
- `cmap` : A colormap (like `'viridis'`) to use if `c` is a number (like `pclass`).
- `alpha` : Transparency (e.g., `alpha=0.5`), which is very useful when many dots overlap.
- `marker` : The shape of the marker (e.g., `marker='x'` or `marker='+'`).

4. Line Plot

- **What is it?** A plot that connects data points with a line.
- **When to use it?** Classically used to show a **trend over a continuous sequence**, most often **time**. The Titanic dataset doesn't have a great time variable, but it's perfect for things like stock prices over a year or temperature over a day.

Titanic Example 07: Since we don't have time, let's (a bit artificially) plot the **count of passengers by age** . *Note: A bar chart is probably better here, but this shows how `ax.plot()` works.*

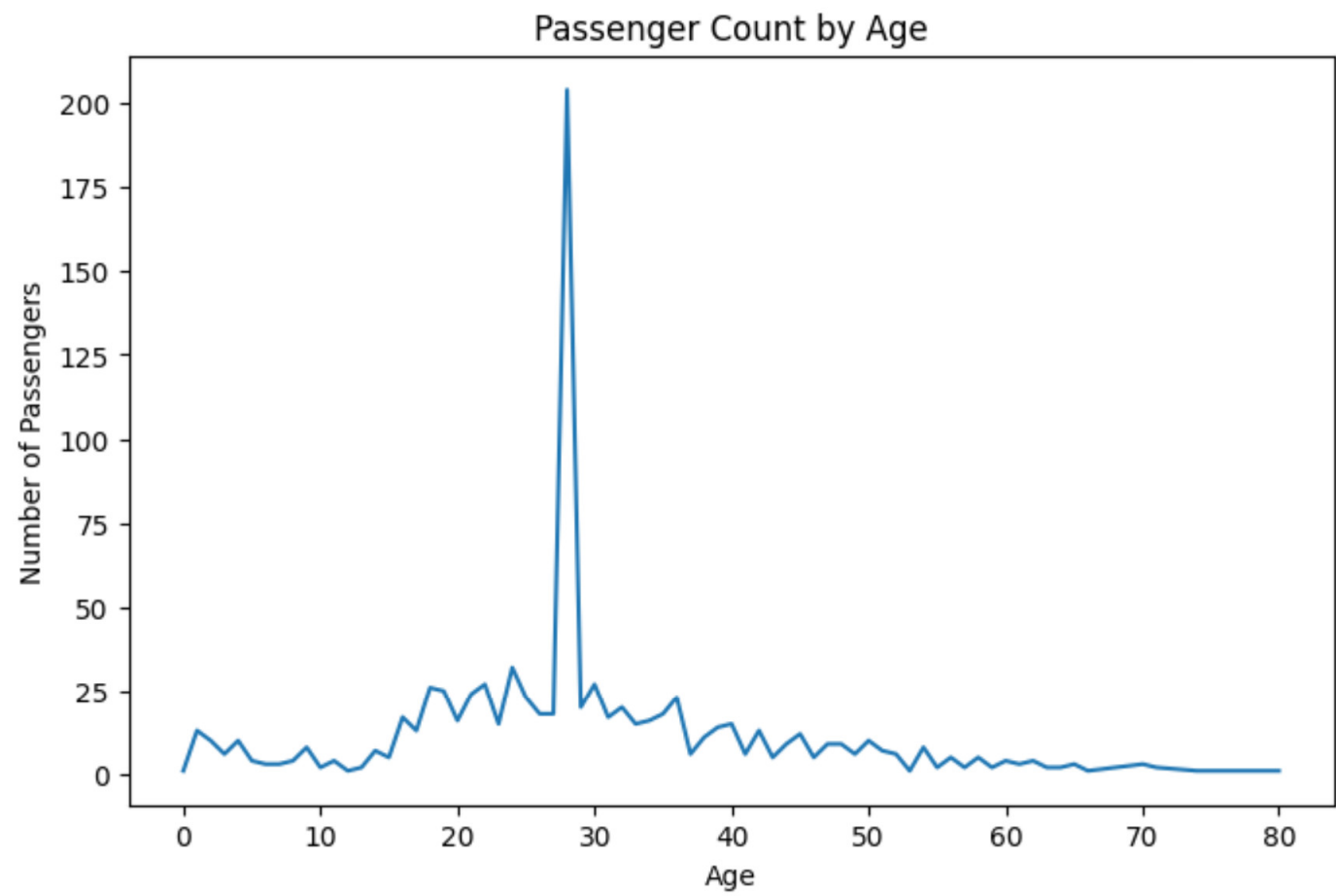
```
In [95]: fig, ax = plt.subplots(figsize=(8, 5))

ageCounts = df['Age'].round().value_counts().sort_index()

ax.plot(ageCounts.index, ageCounts.values)

ax.set_title('Passenger Count by Age')
ax.set_xlabel('Age')
ax.set_ylabel('Number of Passengers')

Out[95]: Text(0, 0.5, 'Number of Passengers')
```

Common Parameters for `ax.plot()` :

- `x` , `y` : The data points. `x` must be in order for the line to make sense.
- `color` : The color of the line (e.g., `color='red'`).
- `linestyle` : How the line is drawn:
 - `'-'` (solid, default)
 - `'--'` (dashed)
 - `'.'` (dotted)
 - `'-.'` (dash-dot)
- `linewidth` : The thickness of the line (e.g., `linewidth=3`).
- `marker` : A marker for each data point (e.g., `marker='o'` for circles).
- `label` : A name for the legend (e.g., `label='Passengers'`). Call `ax.legend()` to show it.

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Matplotlib Cheat Sheet (for `ax` methods)

Plot Function	What It's For	Titanic Example
<code>ax.bar(x, height)</code>	Comparing a quantity across categories .	Number of survivors in each <code>pclass</code> .
<code>ax.hist(x)</code>	Distribution (frequency) of one continuous variable.	The <code>age</code> distribution of all passengers.
<code>ax.scatter(x, y)</code>	Relationship between two continuous variables.	<code>age</code> vs. <code>fare</code> .
<code>ax.plot(x, y)</code>	Trends over a continuous (ordered) sequence.	Stock price over time (or passenger count by <i>sorted</i> age).
<code>ax.boxplot(x)</code>	Distribution of a variable across categories .	<code>fare</code> distribution for each <code>pclass</code> .

Common Customization Commands (call these on `ax`):

- `ax.set_title("My Title")` : Sets the plot's main title.
- `ax.set_xlabel("X-Axis Label")` : Sets the label for the x-axis.
- `ax.set_ylabel("Y-Axis Label")` : Sets the label for the y-axis.
- `ax.legend()` : Displays the legend (requires `label=...` in your plot command).
- `ax.grid(True)` : Adds a background grid.
- `ax.set_xticklabels([...])` : Sets custom text labels for x-axis ticks.
- `ax.set_xticks([...])` : Sets the *location* of x-axis ticks.